

Time: 3Hrs.

Max.
Marks: 60

Note: All questions are compulsory. Attempt any two parts from each question.

Q.1(I) Discuss about Series resonance in detail. A coil having an inductance of 50mH and resistance of 10 ohm is connected in series with a 25 micro farad capacitor across a 200 V ac supply. Calculate (1) resonance frequency of the circuit (2) current flowing at resonance (3) Value of Q_0 .

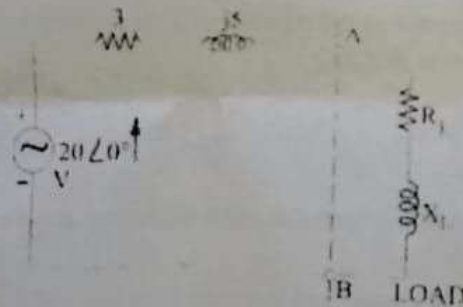
3+3

(II) Explain the two-wattmeter method in detail for a balanced 3 phase star or delta connected load along with its necessary circuit, derivation and phasor diagrams. A 3 phase, 500 V motor load has a power factor of 0.4. Two wattmeter's connected to measure the power shown at the input to be 30 KW. Find reading on each instrument.

3+3

(III) In circuit shown below, which load impedance of p.f. = 0.8 lagging when connected across terminals A and B will draw the maximum power from the source. Also find the power developed in the load and the power loss in the source.

6



Q.2(I) Using nodal analysis, find voltage v in the network shown below. (Fig. 1)

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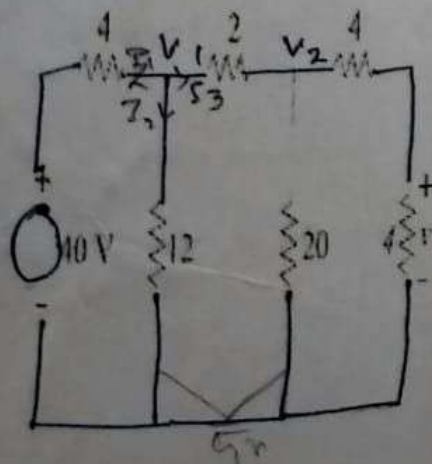


Fig. 1

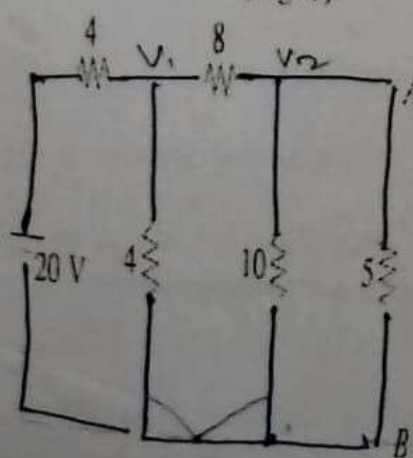
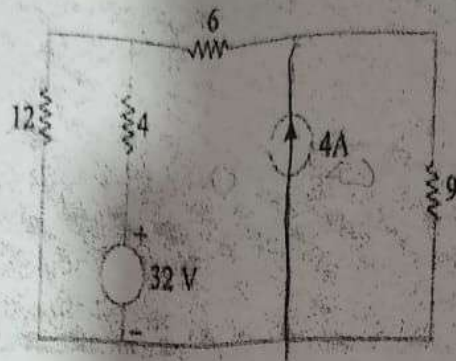


Fig. 2

(II) State Thevenin theorem. Use Thevenin's theorem to calculate the current in the branch AB. (Fig. 2)

6

- (III) Compute the voltage in the 6 ohm resistor in the figure below by applying Superposition Theorem. 6



- Q.3 (I) Compare electric and magnetic circuits. Establish the relationship between magnetomotive force, magnetic flux and magnetic reluctance. 6

- (II) What is transformer? Derive an EMF equation of a single-phase transformer. 6

- (III) A 30KVA, $\frac{2400}{120}$ V, 50Hz single phase transformer has high voltage winding resistance of 0.1Ω and leakage reactance of 0.22Ω . The low voltage winding resistance and leakage reactance are 0.035Ω and 0.012Ω respectively. Calculate
- Equivalent resistance referred to both primary and secondary side
 - Equivalent reactance referred to both primary and secondary side
 - Equivalent impedance referred to both primary and secondary side
 - Copper loss at full load

- Q.4 (I) With a neat circuit diagram explain the construction and working principle of DC Motor. 6

- (II) Derive emf equation of generator and torque equation of motor. 6

- (III) Explain the construction and working principle of generator with the help of neat diagram. 6

- Q.5 (I) State starting methods of single phase induction motor and explain any one of them in details. 6

- (II) Explain construction and working principle of synchronous motor. 6

- (III) With a neat circuit diagram explain the construction and working principle of three phase induction motor. 6